

(Title Page)

Fluid Flow in Pipes

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Executive summary

This experiment was conducted to experimentally determine the flow coefficient (Cf), the friction factor, and the loss coefficient. The flow coefficient was determined at various water flow rates. Likewise, friction head loss was also determined at varying flow rates passing through different materials and sizes. Additionally, the loss coefficient (K) was determined for different flow rates passing through bends, expansions, elbows, and contractions. For the flow coefficient, there was a direct correlation between flow and the resulting coefficient value. For example, at 7.219LPM, the corresponding Cf value is 6.51, and at 14.438LPM, the corresponding Cf value is 26.23. A similar trend between flow rates and Friction head loss is noticed, with increasing flow rates, no matter the material of the pipe or the size of the pipe, for example, a copper pipe with a diameter of .5 inches at a flow rate of 7.219 LPM was found to have a head loss of $0.000219 \text{ m}^2/\text{s}^2$ compared to when the low rate is 12.9942 LPM the head loss is $0.00374 \text{ m}^2/\text{s}^2$.

Introduction

The objective of this experiment is to determine the flow coefficient, the pipe friction head loss, the pipe fitting head loss, and the significance of this experiment to familiarize us with industry equipment used to determine these values. As for the apparatus used in this experiment, we have valves allowing us to control which pipes have flow, a pump to initiate water flow, a pressure drop reader with a temperature reader to allow us to monitor the flow conditions. The setup allows us to test various pipe materials and diameters, as well as elbows, bends, expansions, and contractions, under controlled flow conditions. This experiment has a direct relevance to industrial applications in Chemical processing and water treatment, where being able to acquire pressure drop, and head loss allows us to properly dictate the correct pump sizing.

Theory

The data being collected is the volumetric flow rate of the water in LPM, the pressure drop, and the water temperature. Using these 3 values, we would be able to calculate the following: the flow coefficient, pipe friction head loss, and the pipe fitting head loss. After calibration, we first calculate the flow friction at different flow rates. We do this by using the following equation.

$$Q = C_f A_0 \sqrt{\frac{2 \Delta p}{\rho}} \quad (eq 1)$$

Where Q is the volumetric flow rate, Cf is the flow coefficient, delta p is the differential pressure, and rho is the density of the fluid. Once we are done with this experiment, we move on to the next experiment, where we need to calculate the pipe friction head loss, and the equations used to get this value are the following

$$f = \frac{\Delta P D}{2 L \rho V^2} \quad (eq 2)$$

$$h_L = \frac{\Delta p}{\rho} = f \frac{L}{D} \frac{V^2}{2} \quad (eq 3)$$

$$f = \frac{64}{\text{Re}} \quad (eq 4)$$

$$\Re = \frac{\rho VL}{\mu} \text{ (eq 5)}$$

The reason why we have multiple ways to calculate the friction factor and Reynolds number is to compare our experimental value with our actual value of Reynolds and the friction factor. Once we calculate head loss for this experiment, we can move forward to the next experiment. In the final experiment of this lab, we need to calculate the pipe fitting head loss for the flowing fittings, which are bends, elbows, and contractions. The governing equation used in this experiment is equation number 3. We need to calculate the friction factor as a function of Reynolds number.

Additionally, an orifice meter measures the flow rate of fluids (liquids, gases, or steam) in a pipe by creating a pressure drop when the fluid passes through a precisely sized hole. The concentric orifice plate is recommended for clean liquids, gases, and steam flows when Re ranges from 20,000 and 107 in pipes under 6 inches.

Friction factor is the ratio of shear stress on the bounding walls of the channel to the dynamic pressure of the coolant. The Darcy friction factor and Fanning friction factor is used to calculate pressure loss due to friction in a pipe flow, but they are defined differently. The Darcy factor is for using a different pressure and kinetic energy term in its definition. Although they are both used to calculate head loss due to friction in pipes and ducts.

Results

After calibration, the first experiment was to calculate Cf and to see if this value varies with flow rate.

Orifice						
Flowrate (SCFM)	Pressure (PSI)	Temperature (F)	Flowrate (PSI)	Pressure (PSI)	LPM	Cf
						6.5177248
10	0.21	80.7		25	7.219	29
						9.6547850
12	0.32	80.9		25	8.6628	42
						13.357361
14	0.45	81		24	10.1066	09
						16.876706
16	0.55	81		23	11.5504	63
						22.610292
18	0.78	81		23	12.9942	47
						26.225624
20	0.85	81		22	14.438	11

Table 1: Cf at various flow rates

Here, we can see in Figure 1 that by increasing the flow rate, C_f also goes up which is expected from looking at equation 1.

In the following experiment, we were tasked with finding the friction factors as a function of Reynolds number and comparing them to published data, And to also find the head loss . We were also tasked with answering the following questions: Can this study be used for both laminar and turbulent flow? And what is the difference between the Fanning friction factor and the Darcy friction factor?

The following data results is for PVC pipes

PVC Pipes

0.375"						
Flow Rates (SCFM)	Fanning f	Re	Darcy f	hL	Re (from Fanning)	
			3.09051E-0			
10	0.005956513	2070854.449	5	0.951025229	10744.54045	
			2.57543E-0			
12	0.008122518	2485025.339	5	1.867467723	7879.329666	
			2.37732E-0			
13	0.006920962	2692110.784	5	1.867467723	9247.268844	
			2.20751E-0			
14	0.006796393	2899196.229	5	2.12683824	9416.759845	
			2.06034E-0			
15	0.006498015	3106281.673	5	2.334334654	9849.162083	
			1.93157E-0			
16	0.006980289	3313367.118	5	2.853075688	9168.674521	
			1.71695E-0			
18	0.006116958	3727538.008	5	3.164320309	10462.71644	
			1.62659E-0			
19	0.005760013	3934623.453	5	3.319942619	11111.08597	

0.5"						
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Flow Rates (SCFM)	Fanning f	Re	Darcy f	hL	Re (from Fanning)
10			4.57854E-0		
12	0.011092481	1397826.753	5	0.605197873	5769.674324
			4.22634E-0		
13	0.010261717	1514312.316	5	0.657071977	6236.773323
			3.92446E-0		
14	0.010245183	1630797.879	5	0.760820184	6246.838046
			3.66283E-0		
15	0.009330361	1747283.441	5	0.795402919	6859.327491
16	0.008557056	1863769.004	3.4339E-05	0.829985655	7479.207457
			3.05236E-0		
18	0.007747129	2096740.13	5	0.951025229	8261.1246
			2.74712E-0		
20	0.007416116	2329711.255	5	1.123938908	8629.854758

1" Flow Rates (SCFM)	Fanning f	Re	Darcy f	hL	Re (from Fanning)
10	0.087624257	291213.9069	0.00021977	0.103748207	730.3913532
			0.00015697		
14	0.052157296	407699.4696	8	0.121039575	1227.057473
			0.00010988		
20	0.029208086	582427.8138	5	0.138330942	2191.17406
			8.45268E-0		
26	0.025924337	757156.1579	5	0.207496414	2468.722774
			7.32566E-0		
30	0.0227174	873641.7207	5	0.242079149	2817.223791
			6.10471E-0		
36	0.020283393	1048370.065	5	0.311244621	3155.290646
			5.49424E-0		
40	0.019167806	1164855.628	5	0.363118724	3338.9319
			4.67595E-0		
47	0.016527889	1368705.362	5	0.432284195	3872.242798

Tables 2-4: f, hL, and Re of a .375", .5", and 1" PVC pipes

Copper Pipes

0.402"

Flow Rates (SCFM)	Fanning f	Re	Darcy f	hL	Re (from Fanning)
			3.55157E-0		
10	0.007359437	1802021.651	5	0.004005401	8696.317167
11	0.007602724	1982223.816	3.2287E-05	0.004405941	8418.035018
			2.95964E-0		
12	0.00734666	2162425.981	5	0.004806481	8711.441197
			2.73197E-0		
13	0.006441322	2342628.146	5	0.005207021	9935.848572
			2.53683E-0		
14	0.006492701	2522830.311	5	0.005607561	9857.223121
			2.36771E-0		
15	0.00626915	2703032.476	5	0.006008101	10208.72015
			2.21973E-0		
16	0.006528147	2883234.642	5	0.006408641	9803.701408
			2.08916E-0		
17	0.006366295	3063436.807	5	0.006809181	10052.94264

0.666"					
Flow Rates (SCFM)	Fanning f	Re	Darcy f	hL	Re (from Fanning)
			9.74802E-0		
10	0.045925669	656543.7214	5	0.000880849	1393.556194
			8.86183E-0		
11	0.044280948	722198.0935	5	0.000968934	1445.316853
			8.12335E-0		
12	0.039866032	787852.4657	5	0.001057019	1605.376735
			7.49848E-0		
13	0.035100979	853506.8378	5	0.001145104	1823.310943
			6.96287E-0		
14	0.034170884	919161.21	5	0.001233188	1872.939525
			6.49868E-0		
15	0.032318063	984815.5821	5	0.001321273	1980.316697
			6.09251E-0		
16	0.032141988	1050469.954	5	0.001409358	1991.164943
			5.73413E-0		
17	0.029796065	1116124.326	5	0.001497443	2147.934614
			5.41557E-0		
18	0.028939786	1181778.699	5	0.001585528	2211.48836

1.049"

Flow Rates (SCFM)	Fanning f	Re	Darcy f	hL	Re (from Fanning)
			0.00024183		
10	0.092751415	264643.4408	5	0.000225423	690.0164242
			0.00017273		
14	0.056786581	370500.8171	9	0.000315592	1127.026826
			0.00012091		
20	0.032462995	529286.8816	7	0.000450845	1971.475498
			9.30134E-0		
26	0.021952998	688072.9461	5	0.000586099	2915.319392
			8.06116E-0		
30	0.020611426	793930.3224	5	0.000676268	3105.073909
			6.71763E-0		
36	0.017176188	952716.3868	5	0.000811522	3726.088691
			6.04587E-0		
40	0.015072105	1058573.763	5	0.000901691	4246.254918
			5.14542E-0		
47	0.013436149	1243824.172	5	0.001059487	4763.269628

Table 5-7: f, hL, and Re of a 0.402", 0.666", and 1.049" Copper pipes

Steel Pipes

0.375"					
Flow Rates (SCFM)	Fanning f	Re	Darcy f	hL	Re (from Fanning)
10	0.003249007	2070854.449	3.09E-05	0.004934354	19698.32629
11	0.003043147	2277939.894	2.81E-05	0.005427789	21030.86049
12	0.00300834	2485025.339	2.58E-05	0.005921225	21274.19108
13	0.002627402	2692110.784	2.38E-05	0.00641466	24358.66304
14	0.002762761	2899196.229	2.21E-05	0.006908096	23165.23217
15	0.002695473	3106281.673	2.06E-05	0.007401531	23743.51366
16	0.002665201	3313367.118	1.93E-05	0.007894966	24013.19825
17	0.002435819	3520452.563	1.82E-05	0.008388402	26274.53025

0.5"					
Flow Rates (SCFM)	Fanning f	Re	Darcy f	hL	Re (from Fanning)
10	0.010953142	1164850.968	5.49427E-05	0.002081668	5843.072244
11	0.01056088	1281336.065	4.99479E-05	0.002289835	6060.101052

			5		
			4.57855E-0		
12	0.009507936	1397821.162	5	0.002498002	6731.219057
			4.22636E-0		
13	0.008371484	1514306.259	5	0.002706169	7645.000576
			3.92448E-0		
14	0.008149659	1630791.355	5	0.002914335	7853.089313
			3.66284E-0		
15	0.007707766	1747276.452	5	0.003122502	8303.313827
			3.43392E-0		
16	0.007665773	1863761.549	5	0.003330669	8348.799267
			3.23192E-0		
17	0.007106277	1980246.646	5	0.003538836	9006.122334
			3.05237E-0		
18	0.006902057	2096731.743	5	0.003747003	9272.59801

1"					
Flow Rates (SCFM)	Fanning f	Re	Darcy f	hL	Re (from Fanning)
			2.73687E-0		
10	0.001132433	2338438.319	5	0.002089474	56515.4848
			2.74028E-0		
14	0.001362311	2335526.191	5	0.002086872	46978.99378
			0.00002743		
20	0.001593333	2332614.064	7	0.00208427	40167.37242
			0.00002743		
26	0.001820953	2332614.064	7	0.00208427	35146.43157
			2.74713E-0		
30	0.002281885	2329701.936	5	0.002081668	28046.98747
			2.75057E-0		
36	0.00274512	2326789.809	5	0.002079066	23314.0992
			2.74713E-0		
40	0.00296645	2329701.936	5	0.002081668	21574.60938
			2.74713E-0		
47	0.003651015	2329701.936	5	0.002081668	17529.37197

Table 8-10: f, hL, and Re of a 0.375", 0.5", and 1" Steel pipes

For the final experiment, we were asked to measure Reynolds number, friction factor, loss coefficient, and the head loss for different pipe fittings across various flow rates. The results are as follows:

U-bend				
Flow Rate (LPM)	Re	f	K	hL
		116794.79	12.713030	0.4149928
10.1066	0.00054797	87	99	27
	0.00078281	81756.359	6.2293851	0.4149928
14.438	4	07	87	27
	0.00083872	76305.935	4.6512742	0.4841582
18.0475	9	13	73	99
	0.00072801	87910.063	4.3214604	0.6916547
22.3789	7	51	14	12
	0.00076858	83270.365	3.5248578	0.7608201
25.9884	1	72	73	84
	0.00085397	74943.329	2.8551348	0.7608201
28.876	9	15	77	84
	0.00075485	84784.372	2.8711569	0.9683165
32.4855	6	37	59	97
	0.00073584	86974.850	2.8200023	1.0374820
33.9293	5	07	48	69

Table 11: Re, f, K, and hL of a U-bend

Elbow				
Flow Rate (LPM)	Re	f	K	hL
	0.00136992	46717.919	3.7161056	0.0691654
7.219	4	47	41	71
	0.00205488	31145.279	1.6516025	0.0691654
10.8285	6	65	07	71
	0.00091328	70076.879	2.7870792	0.2074964
14.438	3	2	31	14
	0.00114160	56061.503	1.7837307	0.2074964
18.0475	3	36	08	14
20.9351	0.00079455	80548.137	2.2093374	0.3458273

	6	01	8	56
	0.00095894	66739.884	1.5167778	0.3458273
25.2665	7	95	13	56
	0.00078281	81756.359	1.6257962	0.4841582
28.876	4	07	18	99
	0.00064386	99399.828	1.6822569	0.6916547
33.9293	4	65	68	12

Table 12: Re, f, K, and hL of an elbow bend

90-degree bend				
Flow Rate (LPM)	Re	f	K	hL
	1.08236E-0	59129826	13.310976	0.3458273
10.8285	8	77	62	56
		70955792	11.979878	0.5533237
14.438	9.0197E-09	13	95	7
	9.62101E-0	66521055	9.3592804	0.6224892
17.3256	9	12	33	41
	1.08236E-0	59129826	6.6554883	0.6916547
21.657	8	77	08	12
	9.43599E-0	67825389	6.7360911	0.8991511
24.5446	9	53	07	26
	8.79421E-0	72775171	6.3010540	1.1066475
28.1541	9	41	19	4
	8.81926E-0	72568423	5.5691586	1.2449784
31.7636	9	77	05	82
	8.47852E-0	75484885	5.4232136	1.3833094
33.9293	9	24	5	25

Table 13: Re, f, K, and hL of a 90-degree bend

Contraction				
Flow Rate (LPM)	Re	f	K	hL
		67685.484	33.058862	0.3458273
7.219	0.00094555	24	04	56
	0.00118193	54148.387	17.631393	0.4149928
10.8285	7	39	09	27
	0.00135078	47379.838	11.570601	0.4841582
14.438	6	96	72	99
	0.00168848	37903.871	7.4051850	0.4841582
18.0475	2	17	98	99
	0.00177290	36098.924	5.8771310	0.5533237
21.657	6	93	3	7
		30941.935	4.3178921	0.5533237
25.2665	0.00206839	65	85	7
		33842.742	4.1323577	0.6916547
28.876	0.0018911	12	55	12
	0.00222204	28802.333	2.9931065	0.6916547
33.9293	2	72	68	12

Table 14: Re, f, K, and hL of a contraction

Expansion				
Flow Rate (LPM)	Re	f	K	hL
		247180.918	265.787101	0.27666188
7.219	0.00025892	2	3	5
	0.00031070	205984.098	147.659500	0.34582735
10.8285	4	5	7	6
14.438	0.00034522	185385.688	99.6701629	0.41499282

	6	7	9	7
		197744.734	85.0518724	
18.0475	0.00032365	6	2	0.55332377
	0.00031070	205984.098	73.8297503	0.69165471
21.657	4	5	6	2
	0.00032953	194213.578	59.6664921	0.76082018
25.2665	4	6	3	4
		200834.496	53.9880049	0.89915112
28.876	0.00031867	1	5	6
	0.00030379	210665.555	51.4825222	1.03748206
31.7636	9	3	1	9

Table 15: Re, f, K, and hL of an expansion

Significance of table 1: This table displays the proportional relationship between Cf and flow rate.

Significance of tables 2-10: These tables present how the hL, Re, and f all pivot on the flow rate and pipe diameter. Additionally, the effects of materials are shown when comparing different materials to similar flow rates.

Significance of tables 11-15: These tables lay out how Re, f, k, and hL all change across various pipe fittings and flow rates.

These results were obtained by completing calculations using experimental values. For experiment one, the conversion factor was taken to be the slope of the LPM vs SCFM plot. For experiment two, SCFM was converted to LPM. Then, the flow coefficient, Cf, is found by using (eq1) and isolating it. For experiment three, a rigorous series of conversions is performed to ensure unit cancelation, since f, hL, and Re are unitless. Flow rate goes from SCFM to LPM to m³/s, pressure goes from PSI to N/m², and diameter and length go from inches to m. Additionally, all further constants introduced are in SI units. Then the velocity, in m/s, is

calculated by using the equation: $Q = A \times V$. In other terms, $V = \frac{4Q}{\pi D^2}$. Afterwards, the Fanning

friction factor, f, is calculated using (eq2). However, an assumption is made; density of water, ρ , is a constant 996.85 kg/m³. Subsequently, the head loss, hL, and Reynolds number, Re, are calculated by using (eq3) and (eq4) respectively. However, another assumption is needed for Re; μ is a constant 0.0008671 N*s/m². Lastly, the Fanning friction factor, also f, is found by using (eq 4). For experiment 4, a similar set of rigorous conversions are done. Additionally, the velocity is found the same way. The Reynolds number and Fanning friction factor are found using (eq4) and

(eq2) respectively. The loss coefficient, K , is found as such: $K = f \frac{L}{D}$. Then, the head loss, or h_L is found using: $h_L = K \frac{V^2}{2}$.

As for the associated quantitative uncertainty, since no replicate trials were done, uncertainties are based on instrument specifications, calibration uncertainty, and human error.

Discussion

For the experiment around the orifice, a correlation can be seen that as the flow rate increases, the flow coefficient, C_f , also increases. This can be deduced because the equation used to calculate the flow coefficient uses differential pressure. A basic property of fluid flow is that as the flow rate increases, the pressure drop should also increase.

For the pipe friction head loss experiment, we can see from the calculated results that the Darcy Friction Factor is a much smaller value when compared to the Fanning Friction Factor. Fanning friction factor is determined using (eq2), and Darcy Friction Factor is determined using (eq4), and solving for f with Reynolds' number. For Darcy and Fanning, we can see a correlation of a decreasing f as the flow rate is increased. Moreover, for the calculation of Reynolds' number, 2 separate ways were done. One uses the Fanning Friction Factor to solve Reynolds' number, and the other uses (eq5). Although the difference in calculations, the correlation remains the same as the friction factor decreases, the Reynolds' number increases, allowing for an inverse relationship. As for the question, whether this study can cover laminar and turbulent flow. Based on the results and properties of Transport Phenomena, a Re value greater than 4000 is turbulent and less than 2300 is laminar. The results show more values that are greater than 4000, but there are values that are less than 2300. So, it seems this study can partially cover laminar flow, but most of the study will cover turbulent flow. The answer to this question is a bit decisive due to the limitations of piping sizes. The largest pipe showed more of a laminar flow, and due to the largest pipe being limited to 1 inch, there is more to be tested. The next steps would be to allow a larger pipe to be experimented on to determine if the larger pipes allow for a more laminar flow. From a Transport Phenomena view, a larger pipe would allow more flow to be laminar due to the pressure drop being much smaller. With a smaller pressure drop, this allows the fluid to flow due to the fluid being smoother in transport, with less mixing and turbulence.

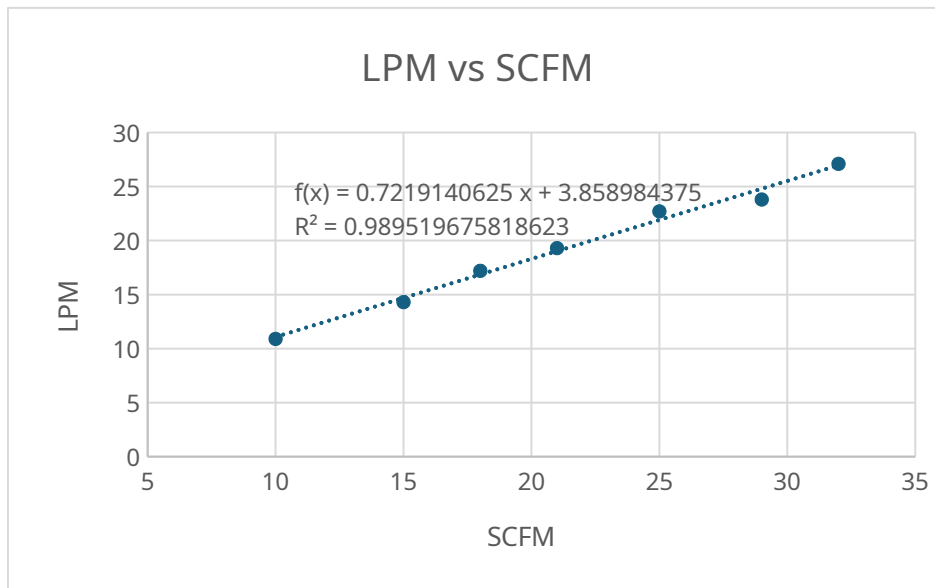
For the Pipe Fitting Head Loss experiment, we were able to find the equivalent length of diameters to be 7.10 inches for the U-bend fitting, 4.12 inches for the elbow fitting, 8.42 inches for the 90-degree fitting, 14.82 inches for the contraction fitting, and 6.8 inches for the expansion fitting. The equivalent length of diameters relates the straight length of a pipe fitting to the diameter of the pipe. This allows the length of the pipe fitting to be calculated and relates the length to the frictional loss from the specific fitting. This is important in chemical engineering when you want to find the friction loss and pressure drop for a series of pipes and pipe fittings, so your pump is designed correctly. We were able to estimate these values by calculating the loss coefficient, K , using the equation $\Delta P = \frac{K \times \rho \times V^2}{2}$ by solving for K as the rest of the variables are known. Once the loss coefficient values are calculated for each pipe fitting, it is then

multiplied by the diameter of the pipe to obtain an “equivalent length of diameters.” Moreover, the head loss coefficient was able to be calculated using the equation $h_L = \frac{K \times V^2}{2}$, using this equation after the loss coefficients were found.

Appendix I: Data Tabulation/Graphs

Experiment 1: Rotameter Calibration

SCFM	LPM
	10.953
10	35
	11.217
16	39
	11.742
26	42
	11.506
32	02



Experiment 2: Orifice Meter

Orifice					
Flowrate (SCFM)	Pressure (PSI) (dP)	Temperature (F)	Flowrate Pressure (PSI)	LPM	Cf
					6.5177248
10	0.21	80.7	25	7.219	29
					9.6547850
12	0.32	80.9	25	8.6628	42

14	0.45	81	24	10.1066	13.35736109
16	0.55	81	23	11.5504	16.87670663
18	0.78	81	23	12.9942	22.61029247
20	0.85	81	22	14.438	26.22562411

Experiment 3: Friction Head Loss

.375" PVC						
Flow Rates (SCFM)	Pressure (PSI)	Temperature (F)	Flow rate (LPM)	Flow rate (m ³ /s)	Velocity (m/s)	Pressure (N/m ²)
10	0.55	78.9	7.219	0.000120317	1.688518959	3792.118
12	1.08	78.9	8.6628	0.00014438	2.026222751	7446.3408
13	1.08	78.9	9.3847	0.000156412	2.195074647	7446.3408
14	1.23	78.9	10.107	0.000168443	2.363926543	8480.5548
15	1.35	78.9	10.829	0.000180475	2.532778439	9307.926
16	1.65	78.9	11.55	0.000192507	2.701630335	11376.354
18	1.83	78.9	12.994	0.00021657	3.039334126	12617.4108
19	1.92	78.9	13.716	0.000228602	3.208186022	13237.9392

0.5" PVC						
Flow Rates (SCFM)	Pressure (PSI)	Temperature (F)	Flow rate (LPM)	Flow rate (m ³ /s)	Velocity (m/s)	Pressure (N/m ²)
10	0.26	79.2	7.219	0.000120317	0.949791915	1792.6376
12	0.35	79.3	8.6628	0.00014438	1.139750297	2413.166

Fluid Flow in Pipes

University of Illinois

13	0.38	79.4	9.3847	0.0001564	1.2347294	
				12	89	2620.0088
14	0.44	79.3	10.107	0.0001684	1.3297086	
				43	8	3033.6944
15	0.46	79.3	10.829	0.0001804	1.4246878	
				75	72	3171.5896
16	0.48	79.2	11.55	0.0001925	1.5196670	
				07	63	3309.4848
18	0.55	79.2	12.994	0.0002165	1.7096254	
				7	46	3792.118
20	0.65	79.2	14.438	0.0002406	1.8995838	
				33	29	4481.594

1" PVC						
Flow Rates (SCFM)	Pressure (PSI)	Temperature (F)	Flow rate (LPM)	Flow rate (m ³ /s)	Velocity (m/s)	Pressure (N/m ²)
				0.0001203	0.2374479	
10	0.06	79.6	7.219	17	79	413.6856
				0.0001684	0.3324271	
14	0.07	79.5	10.107	43	7	482.6332
				0.0002406	0.4748959	
20	0.08	79.5	14.438	33	57	551.5808
				0.0003128	0.6173647	
26	0.12	79.6	18.769	23	44	827.3712
				0.0003609	0.7123439	
30	0.14	79.7	21.657	5	36	965.2664
				0.0004331	0.8548127	
36	0.18	79.4	25.988	4	23	1241.0568
				0.0004812	0.9497919	
40	0.21	79.4	28.876	67	15	1447.8996
				0.0005654		
47	0.25	79.4	33.929	88	1.1160055	1723.69

0.402" Copper						
Flow Rates (SCFM)	Pressure (PSI)	Temperature (F)	Flow rate (LPM)	Flow rate (m ³ /s)	Velocity (m/s)	Pressure (N/m ²)
10	0.48	80.5	7.219	0.0001203	1.4693199	3309.4848

Fluid Flow in Pipes

University of Illinois

				17	34	
				0.0001323	1.6162519	
11	0.6	80.1	7.9409	48	27	4136.856
				0.0001443	1.7631839	
12	0.69	79.9	8.6628	8	21	4757.3844
				0.0001564	1.9101159	
13	0.71	79.9	9.3847	12	14	4895.2796
				0.0001684	2.0570479	
14	0.83	79.8	10.107	43	08	5722.6508
				0.0001804	2.2039799	
15	0.92	79.8	10.829	75	01	6343.1792
				0.0001925	2.3509118	
16	1.09	79.9	11.55	07	95	7515.2884
				0.0002045	2.4978438	
17	1.2	79.8	12.272	38	88	8273.712

0.666" Copper						
Flow Rates (SCFM)	Pressure (PSI)	Temperature (F)	Flow rate (LPM)	Flow rate (m^3/s)	Velocity (m/s)	Pressure (N/m^2)
				0.0001203	0.5353280	
10	0.24	80.2	7.219	17	73	1654.7424
				0.0001323	0.5888608	
11	0.28	80.4	7.9409	48	8	1930.5328
				0.0001443	0.6423936	
12	0.3	80.2	8.6628	8	87	2068.428
				0.0001564	0.6959264	
13	0.31	80.1	9.3847	12	95	2137.3756
				0.0001684	0.7494593	
14	0.35	80	10.107	43	02	2413.166
				0.0001804	0.8029921	
15	0.38	80	10.829	75	09	2620.0088
				0.0001925	0.8565249	
16	0.43	80	11.55	07	16	2964.7468
				0.0002045	0.9100577	
17	0.45	80	12.272	38	24	3102.642
18	0.49	79.9	12.994	0.0002165	0.9635905	3378.4324

1.049" Copper						
Flow Rates (SCFM)	Pressure (PSI)	Temperature (F)	Flow rate (LPM)	Flow rate (m ³ /s)	Velocity (m/s)	Pressure (N/m ²)
				0.0001203	0.2157831	
10	0.05	80.3	7.219	17	36	344.738
				0.0001684	0.3020963	
14	0.06	80.2	10.107	43	9	413.6856
				0.0002406	0.4315662	
20	0.07	80.1	14.438	33	72	482.6332
				0.0003128	0.5610361	
26	0.08	80.1	18.769	23	54	551.5808
				0.0003609	0.6473494	
30	0.1	80	21.657	5	08	689.476
				0.0004331	0.7768192	
36	0.12	79.9	25.988	4	9	827.3712
				0.0004812	0.8631325	
40	0.13	80	28.876	67	44	896.3188
				0.0005654	1.0141807	
47	0.16	80	33.929	88	39	1103.1616

0.375" Steel						
Flow Rates (SCFM)	Pressure (PSI)	Temperature (F)	Flow rate (LPM)	Flow rate (m ³ /s)	Velocity (m/s)	Pressure (N/m ²)
				0.0001203	1.6885189	
10	0.3	80.1	7.219	17	59	2068.428
				0.0001323	1.8573708	
11	0.34	80.5	7.9409	48	55	2344.2184
				0.0001443	2.0262227	
12	0.4	80.3	8.6628	8	51	2757.904
				0.0001564	2.1950746	
13	0.41	80.3	9.3847	12	47	2826.8516
				0.0001684	2.3639265	
14	0.5	80.3	10.107	43	43	3447.38
				0.0001804	2.5327784	
15	0.56	80.3	10.829	75	39	3861.0656
16	0.63	80.3	11.55	0.0001925	2.7016303	4343.6988

Fluid Flow in Pipes

University of Illinois

				07	35	
				0.0002045	2.8704822	
17	0.65	80.3	12.272	38	31	4481.594

0.5" Steel						
Flow Rates (SCFM)	Pressure (PSI)	Temperature (F)	Flow rate (LPM)	Flow rate (m ³ /s)	Velocity (m/s)	Pressure (N/m ²)
				0.0001203	0.9497881	
10	0.24	80.2	7.219	17	15	1654.7424
				0.0001323	1.0447669	
11	0.28	80.4	7.9409	48	27	1930.5328
				0.0001443	1.1397457	
12	0.3	80.2	8.6628	8	38	2068.428
				0.0001564	1.2347245	
13	0.31	80.1	9.3847	12	5	2137.3756
				0.0001684	1.3297033	
14	0.35	80	10.107	43	61	2413.166
				0.0001804	1.4246821	
15	0.38	80	10.829	75	73	2620.0088
				0.0001925	1.5196609	
16	0.43	80	11.55	07	85	2964.7468
				0.0002045	1.6146397	
17	0.45	80	12.272	38	96	3102.642
				0.0002165	1.7096186	
18	0.49	79.9	12.994	7	08	3378.4324

1" Steel						
Flow Rates (SCFM)	Pressure (PSI)	Temperature (F)	Flow rate (LPM)	Flow rate (m ³ /s)	Velocity (m/s)	Pressure (N/m ²)
				0.0009661	1.9066996	
10	0.05	80.3	57.969	43	42	344.735
				0.0009649	1.9043251	
14	0.06	80.2	57.896	4	71	413.682
				0.0009637	1.9019507	
20	0.07	80.1	57.824	37	01	482.629
				0.0009637	1.9019507	
26	0.08	80.1	57.824	37	01	551.576
30	0.1	80	57.752	0.0009625	1.8995762	689.47

				33	31	
				0.0009613	1.8972017	
36	0.12	79.9	57.68	3	6	827.364
				0.0009625	1.8995762	
40	0.13	80	57.752	33	31	896.311
				0.0009625	1.8995762	
47	0.16	80	57.752	33	31	1103.152

Experiment 3: Pipe Fitting Head Loss

U-bend Pressure Drop (PSI)	Temperat ure (F)	Flow Rate (LPM)	Pressure (N/m ²)	Flow rate (m ³ /s)	Velocity (m/s)
0.06	80.9	10.1066	413.6856	0.000168443	0.25551167
0.06	81	14.438	413.6856	0.000240633	0.365016671
0.07	80.9	18.0475	482.6332	0.000300792	0.456270839
0.1	80.8	22.3789	689.476	0.000372982	0.56577584
0.11	80.7	25.9884	758.4236	0.00043314	0.657030008
0.11	80.8	28.876	758.4236	0.000481267	0.730033342
0.14	80.8	32.4855	965.2664	0.000541425	0.82128751
0.15	80.8	33.9293	1034.214	0.000565488	0.857789177

Elbow Pressure Drop (PSI)	Temperat ure	Flow Rate (LPM)	Pressure (N/m ²)	Flow rate (m ³ /s)	Velocity (m/s)
0.01	80.9	7.219	68.9476	0.000120317	0.192937054
0.01	80.9	10.8285	68.9476	0.000180475	0.28940558
0.03	81	14.438	206.8428	0.000240633	0.385874107
0.03	80.9	18.0475	206.8428	0.000300792	0.482342634
0.05	80.9	20.9351	344.738	0.000348918	0.559517456
0.05	80.9	25.2665	344.738	0.000421108	0.675279688
0.07	80.9	28.876	482.6332	0.000481267	0.771748215
0.1	80.8	33.9293	689.476	0.000565488	0.906804152

90 degree bend Pressure Drop (PSI)	Temperat ure	Flow Rate (LPM)	Pressure (N/m ²)	Flow rate (m ³ /s)	Velocity (m/s)
0.05	81	10.8285	344.738	0.000180475	0.227950059
0.08	81.2	14.438	551.5808	0.000240633	0.303933413
0.09	81.1	17.3256	620.5284	0.00028876	0.364720095
0.1	81	21.657	689.476	0.00036095	0.455900119
0.13	80.9	24.5446	896.3188	0.000409077	0.516686801

0.16	81	28.1541	1103.1616	0.000469235	0.592670155
0.18	80.9	31.7636	1241.0568	0.000529393	0.668653508
0.2	81	33.9293	1378.952	0.000565488	0.71424352

Contraction					
Pressure Drop (PSI)	Temperature	Flow Rate (LPM)	Pressure (N/m ²)	Flow rate (m ³ /s)	Velocity (m/s)
0.05	81.4	7.219	344.738	0.000120317	0.144644099
0.06	81.4	10.8285	413.6856	0.000180475	0.216966148
0.07	81.4	14.438	482.6332	0.000240633	0.289288198
0.07	81.3	18.0475	482.6332	0.000300792	0.361610247
0.08	81.2	21.657	551.5808	0.00036095	0.433932296
0.08	81.2	25.2665	551.5808	0.000421108	0.506254346
0.1	81.2	28.876	689.476	0.000481267	0.578576395
0.1	81.2	33.9293	689.476	0.000565488	0.679827264

Expansion					
Pressure Drop (PSI)	Temperature	Flow Rate (LPM)	Pressure (N/m ²)	Flow rate (m ³ /s)	Velocity (m/s)
0.04	81.1	7.219	275.7904	0.000120317	0.045627084
0.05	81.1	10.8285	344.738	0.000180475	0.068440626
0.06	81.3	14.438	413.6856	0.000240633	0.091254168
0.08	81.3	18.0475	551.5808	0.000300792	0.11406771
0.1	81.3	21.657	689.476	0.00036095	0.136881252
0.11	81.2	25.2665	758.4236	0.000421108	0.159694794
0.13	81.2	28.876	896.3188	0.000481267	0.182508336
0.15	81.1	31.7636	1034.214	0.000529393	0.200759169

Appendix I: Sample Calculations

For calibration,

$$y = .7219x + 3.859$$

slope, m, is the conversion factor for SCFM to LPM, so,

$$10 \text{ SCFM} \times .7219 = 7.219 \text{ LPM}$$

For determination of flow coefficients,

$$Q = C_f A_0 \sqrt{\frac{2 \Delta p}{\rho}}$$

$$7.219 = C_f \times .7178 \sqrt{\frac{2 \times .21}{1}}$$

$$C_f = 6.5177$$

LPM to m³/s

$$7.219 \text{ LPM} / 60,000 = .00012 \text{ m}^3/\text{s}$$

PSI to N/m²,

$$.55 \text{ PSI} \times 6894.6 = 3792.12 \text{ N/m}^2$$

Velocity calculation,

$$V = \frac{4 \times Q}{\pi \times D^2}$$

$$V = \frac{4 \times .00012}{\pi \times .009525^2}$$

Fanning friction factor,

$$f_F = \frac{P \times D}{2 \times L \times \rho \times D^2}$$

$$f_F = \frac{3,792.12 \times .00012}{2 \times 1.0668 \times .0008671 \times 996.85^2}$$

$$f_F = .00597$$

Reynolds' number,

$$\Re = \frac{\rho V L}{\mu}$$

$$\Re = \frac{996.85 \times 1.69 \times 1.0668}{.000867}$$

$$\Re = 2,070,854$$

Darcy friction factor,

$$f_D = \frac{64}{2,070,854}$$

$$f_D = 3.09\text{E-}05$$

Reynolds' number from the fanning factor,

$$\Re = \frac{64}{f_F}$$

$$\Re = \frac{64}{.00596}$$

$$\Re = 10,744.5$$

Head loss,

$$h_L = f \frac{L}{D} \frac{V^2}{2}$$

$$h_L = .00596 \times \frac{1.0668 \times 1.69^2}{996.85 \times 2}$$

$$h_L = .951$$

Loss coefficient,

$$K = \frac{2 \times P}{\rho \times V^2}$$

$$K = \frac{2 \times 413.69}{996.85 \times .256^2}$$

$$K = 12.71$$

Head Loss using loss coefficient,

$$h_L = \frac{K \times V^2}{2}$$

$$h_L = \frac{12.71 \times .256^2}{2}$$

$$h_L = .415$$

Equivalent length of diameters

$$ELD = K \times D$$

$$ELD = 6.23 \times .029$$

$$ELD = .180 \text{ m} \times 39.37$$

$$ELD = 7.1 \text{ inches}$$

Appendix II: Error Analysis

Some sources of experimental error lie within the pipes themselves. Some pipes would leak water, and although the leakage could be negligible in hindsight, it could cause disruptions in flow and false readings. This error in equipment can be eliminated by regular maintenance and renewal, repair, or replacement of piping. Moreover, the factor by which the flow rate is multiplied, determined from the calibration, can cause inconsistencies in flow rate determination. For example, the in-lab examples had a conversion factor of .8455 while we had a conversion factor of .7219. This can cause differences in results depending on the consistency of the calibration of different groups. Just from the value from the in-lab example to ours lies a 15.77% difference, which can change the values of all calculations by 15.77%. This is just our comparison, and the value can differ largely depending on calibration consistency. A way to prohibit this factor from impeding results, a universal factor can be declared to ensure consistent results. Human error is always a possibility since measurements of pipe diameter, length, and the reading of the flow meter can differ. Additionally, when conducting the experiment, we would forget to purge the streams, and if we had never realized this error, our values could have been skewed greatly.

Appendix III: Individual Team Contributions

NAME: Adil Feratovic

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	
Pre-Lab Editing	2	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	3	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	5	
Executive Summary (final lab)	0	

Introduction (Final lab)	1	
Theory (Final Lab)	1	
Results (final lab)	5	
Discussion (final lab)	1	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	10	
Error Analysis (final lab)	1	
Sample Calculations (final Lab)	0	
Power Point Presentation	1	
Total		

NAME: Hussein Rezk

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	
Pre-Lab Editing	3	
Experimental Objective (Prelab)	3	
Apparatus (prelab)	1	
Materials & Supplies (prelab)	1	
Procedure (prelab)	1	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	1	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	2	
Discussion (final lab)	2	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	1	
Sample Calculations (final	1	

Lab)		
Power Point Presentation	2	
Total	27	

NAME:

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)		
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		

Power Point Presentation		
Total		

NAME: Aaron Casas

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	
Pre-Lab Editing	1	
Experimental Objective (Prelab)	1	
Apparatus (prelab)	1	
Materials & Supplies (prelab)	1	
Procedure (prelab)	1	
Project Safety Evaluation (prelab)		
Final Lab Editing	2	
Executive Summary (final lab)	2	
Introduction (Final lab)	2	
Theory (Final Lab)	3	
Results (final lab)	3	
Discussion (final lab)	2	
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	12	
Error Analysis (final lab)		
Sample Calculations (final Lab)		

Power Point Presentation		
Total		

Appendix IV: References

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Fluid Flow in Pipes Lab Manual

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